A formal knowledge base for amplicon sequencing data and geochemistry of the Rub' al Khali desert

Rund Tawfiq[1], Marwa Abdelhakim[1], Sulaiman M. Alajel[3], Mohammed Alarawi[1], Hind Aldakhil[8], Abderahmane Derouiche[1], Daniela I. Drautz-Moses[4], Michel Dumontier[7], Raik Grünberg[6], Alejandra Lopez-Velazquez[1], Susana Martinez Arbas[2], Kexin Niu[1], Krishnakumar Sivakumar[1], Tiannyu Wang[5], Xiang Zhao[4], Jood Zubair[1], Magnus Rueping[5], Robert Hoehndorf[1*]

[1]Bio-Ontology Research Group (BORG), Computer, Electrical and Mathematical Sciences and Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia.
[2]NIUM, Esch-sur-Alzette, Luxembourg.
[3]Ministry of Environment, Water and Agriculture, Saudi Arabia.
[4]Bioscience Core Lab, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia.
[5]Physical Science and Engineering (PSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia.
[6]Biomedical Sciences Division (BioMed), King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia.
[7]Institute of Data Science, Department of Advanced Computing Sciences, Faculty of Science and Engineering, Maastricht University, Maastricht, The Netherlands.
[8]National Livestock & Fisheries Development Program (NLFDP), Ministry of Environment, Water and Agriculture (MEWA), Riyadh, 13413, Saudi Arabia.

*Corresponding author(s). E-mail(s): robert.hoehndorf@kaust.edu.sa;

Abstract

Integrating microbiome sequence data with environmental and geochemical metadata remains a significant challenge due to the heterogeneity of data formats and a lack of standardized semantic representations. Building such an integrated resource requires addressing several interlocking challenges: defining a unified data model across instruments and observation types, harmonizing taxonomic identifiers across reference databases, validating incoming data against shared schemas, and supporting semantic query answering at scale under a tractable reasoning regime. Here, we present a high-resolution microbiome dataset from five campaigns across a 1,043 km transect of the Empty Quarter (Rub' al Khali) of Saudi Arabia, comprising 2,516 individual samples from 60 primary sampling sites and special-interest locations. A 2,550-row source ledger (2,516 non-control records) connects specimens, DNA extractions, libraries, sequencing runs and community profiles; the canonical feature table contains 1,271 taxonomic profiles, of which 1,237 pass quality filtering. These biological sequences are joined to 274 field environmental logs, 12,936 monthly climate measurements, 712 archived-soil pH observations, and two X-ray fluorescence workflows (71 field sessions and 725 laboratory records). We augment these data with a formal knowledge base expressed in OWL 2 DL using the Semanticscience Integrated Ontology (SIO), with ShEx-based ingest validation and consistency of the tractable modules checked under the OWL 2 EL profile with ELK. This semantic layer enables complex cross-domain queries and provides a comparative baseline for studying microbial adaptation in extreme environments.

Keywords: Knowledge Graph, Metagenomics, Rub' al Khali, SIO, Semantic Web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is the largest contiguous sand desert on Earth, a hyper-arid environment that spans approximately 500,000 km^2 of the southern Arabian Peninsula [1]. The region is characterized by extreme and fluctuating atmospheric temperatures, negligible and highly irregular rainfall [2], and high humidity at the western edge [3]. Despite these abiotic stresses, hyper-arid desert soils harbor diverse microbial communities [4]. Deep knowledge of the biogeochemistry of such extreme environments can predict ecosystem responses to climate change and inform conservation and land-use policy[5].

Saudi Arabia already has a substantial arid soil microbiome literature. Previous studies examined elevation gradients [3], rainfall responses across soil compartments [6], bulk and rhizosphere soils across three regions [7], rhizosphere bacteria of pioneer sand plants near Jizan [8], and bacteria associated with native desert plants [9]. More recent surveys characterized fertility islands and plant-associated networks [10, 11], sandy soils near natural and artificial water points [12], rhizosphere and endosphere communities across northern reserves [13], and Red Sea coastal sabkhas [14]. Fungal studies progressed from a small number of bulk soil samples from the western coast [15, 16] to a standardized single survey of 85 locations across Saudi Arabia [17], although a recent synthesis identified substantial geographic and taxonomic gaps in Arabian biological soil crust research [18]. In the United Arab Emirates, related studies profiled a Rub' al-Khali interdune microbial mat [19] and desert-plant rhizosphere and endosphere communities [20]. These studies establish that a survey of hyperarid soils in Saudi Arabia is not itself new. What remains missing is a repeated, multi-campaign transect in which every community profile, the sample it came from, and the measurements taken alongside it form a connected record that can be queried directly rather than assembled from separate data records.

To characterize complex ecosystems, researchers must integrate heterogeneous and multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that link physical samples to their digital representations. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility.

Engineers originally developed the concept of a digital twin, and environmental scientists increasingly adopt this approach [21, 22]. Recent frameworks, such as TwinEco, standardize the development of dynamic, data-driven digital twins in ecology to avoid fragmented software designs [23]. Additionally, regional natural resource management benefits from artificial intelligence-assisted digital twin architectures that unify heterogeneous environmental observations [24]. Such digital twins show strong predictive accuracy when they incorporate continuous data streams, such as citizen science observations for real-time biodiversity forecasts [25]. To guarantee that these models remain interoperable and reusable across domains, researchers emphasize the integration of datasets, models, and computational workflows through Findable, Accessible, Interoperable, and Reusable (FAIR) principles [26]. However, while crops [27] and specific regions [22] have digital representations, researchers have not yet developed semantic representations of entire desert ecosystems. Digital twins in the plant sciences span cells, organs, individual plants, plots, and fields [28]; ecosystem digital twins, by contrast, model landscapes and environments and leave out the microbial communities within them.

Constructing such a resource for an entire desert ecosystem requires addressing several interlocking challenges that have shaped the design of this work. First, data are produced by heterogeneous instruments and protocols (16S amplicon sequencing, X-ray fluorescence, GPS-tagged field notes, gridded climate records), so a *unified data model* is needed that can represent samples, processes, qualities, and quantities under a common upper-level vocabulary rather than as disconnected tables. Second, taxonomic labels are reported by multiple classifier workflows as text strings, and the same label can denote different taxa across reference databases and releases, so they must be harmonized against a single reference taxonomy without losing the original assertions. Third, because data are deposited in batches over five field campaigns, *schema validation at ingest* is required to catch structural errors before they propagate into the graph. Fourth, the knowledge graph must support *semantic query answering at scale*: with tens of millions of triples, expressive OWL 2 DL reasoning is computationally infeasible, so a tractable reasoning regime must be selected and its trade-offs made explicit. Fifth, a *user-facing query and exploration interface* is required so that domain scientists who are not fluent in SPARQL can interact with the resource. We return to each of these challenges in the usage notes (Section 6) and link them to the corresponding design choices in the methods and knowledge representation sections.

Community reporting standards and knowledge-graph infrastructure define the reuse target this resource is built against. The Tara Oceans consortium set the model for publishing a multi-campaign survey as open registries of campaigns, stations and sampling events, with query routes from every registry entry to the environmental and sequence data it describes [29]; we follow that model at landscape scale and add a semantic layer over the registries. The FAIR principles set the criteria a deposited resource is judged against [30]. MIxS/MIMARKS define contextual fields for marker-gene and environmental sequence data [31], and the STREAMS guideline extends reporting practice to environmental microbiomes [32]. The National Microbiome Data Collaborative shows how persistent identifiers, standardized metadata and linked-data schemas support reuse across environmental microbiome studies [33]. On the knowledge-graph side, KG-Microbe connects microbial taxa with traits, environments and growth preferences [34], and MetagenomicKG integrates taxonomic, functional and biomedical sources for metagenomic applications [35]. These resources establish that a microbiome knowledge graph is not itself novel. The contribution of the present resource is the joined, campaign-level provenance of locations, physical specimens, laboratory processes, geochemistry, climate, sequence runs, quality control and taxon observations from a single landscape survey, rather than vocabulary coverage or graph construction alone.

Here, we describe a versioned, provenance-preserving knowledge graph integrating repeated desert microbiome, environmental, climate, pH, and geochemical observations across the Rub' al-Khali. The knowledge graph is expressed in OWL 2 DL [36] and built from RDF instance data [37]; because reasoning over the full graph in OWL 2 DL is infeasible, its consistency is checked under the OWL 2 EL profile with the ELK reasoner (Section 5). This knowledge graph integrates data from five field campaigns (2023–2025) along a transect of more than 1,000 km across the Saudi Rub' al-Khali, from the western escarpment to the eastern border with Oman. A versioned source ledger of 2,550 rows links field specimens to their downstream products; 2,516 non-control rows are represented as sample individuals in the graph. An amplicon feature table contains 1,271 profiles, of which 1,237 pass quality filtering across 64 numeric site labels. We incorporated X-ray fluorescence (XRF) measurements from two complementary workflows: in-situ handheld field sessions , and processed laboratory records of archived material from all five campaigns. We further integrated site-specific environmental metadata recorded at each visit (274 field logs), 712 admitted archived-soil pH observations, and monthly climate records derived from the Open-Meteo database [38].

We structured the knowledge graph with the Semanticscience Integrated Ontology (SIO) [39] as the upper-level framework. This design allows interoperability with other SIO-based datasets. We modeled domain-specific concepts as subclasses of SIO entities, and all measurements follow SIO's ontology design pattern for measurement processes. We grounded chemical entities in the Chemical Entities of Biological Interest (ChEBI) ontology [40], environmental contexts in the Environment Ontology (ENVO) [41], and taxonomic identifiers in the NCBI Taxonomy [42].

Because it encodes the full provenance chain from physical sample collection to digital data products, the knowledge graph is designed as a FAIR foundation for a future Rub' al-Khali digital twin; building and validating such a twin is beyond the scope of this paper. The graph allows researchers to execute cross-domain queries that otherwise demand complex joins across multiple heterogeneous tables in a conventional database, and it keeps observations, models, and workflows interoperable through shared identifiers and vocabularies [26]. We expect this to support hypothesis generation, intervention simulation, and evidence-based policy in future work.

This resource makes several contributions. On the scientific side, it provides a coordinated multi-trip desert microbiome and geochemistry dataset—16S rRNA amplicon profiles, XRF elemental measurements, archived-soil pH, GPS-tagged field metadata, and monthly climate records—for 2,516 sample individuals from five campaigns across a 1,043 km transect of the Rub' al-Khali, together with a harmonized taxonomic abundance table that maps amplicon sequence variants onto NCBI Taxonomy while preserving the original assertions. On the technical side, the resource comprises an OWL 2 DL knowledge graph built on SIO with domain classes grounded in ChEBI, ENVO, and NCBI Taxonomy, modelling the full provenance chain from field sampling to digital data products. We formalize reusable ontology design patterns for sampling, measurement, taxonomic abundance, and provenance aligned with SIO's core idioms. A ShEx-based ingest validation workflow enforces structural shapes on every RDF batch before loading, and the ELK reasoner is used to check the logical consistency of the tractable modules under the OWL 2 EL profile. A set of competency questions documents the intended query workload and serves as regression tests, and a web portal with an interactive map, per-site dashboards, and a SPARQL editor exposes the knowledge graph to domain scientists without requiring SPARQL fluency. The companion ecological analysis reports the inferential results for the quality-controlled community profiles, laboratory-XRF gradient, rainfall, spatial turnover and association networks [43]; the present paper describes the reusable data products, provenance model, identifier mappings and their executable validation.

2 Methods

2.1 Study area and sampling design

We conducted five sampling campaigns across the Rub' al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E). The 60 ordered sampling sites yield a 1,043-km cumulative path and lie 1,015 km apart end to end. Figure 1 illustrates the altitude profile across this transect.

The transect comprises 60 primary sampling sites at predetermined geographic coordinates, and 10 named sites of special interest (water wells, a hot spring, and an oil spill site). Trip 1 additionally used the numeric identifiers 61–64. Matching their coordinates against the field records shows that these refer to four of the sites of special interest rather than further locations, so we retain the numeric identifiers as

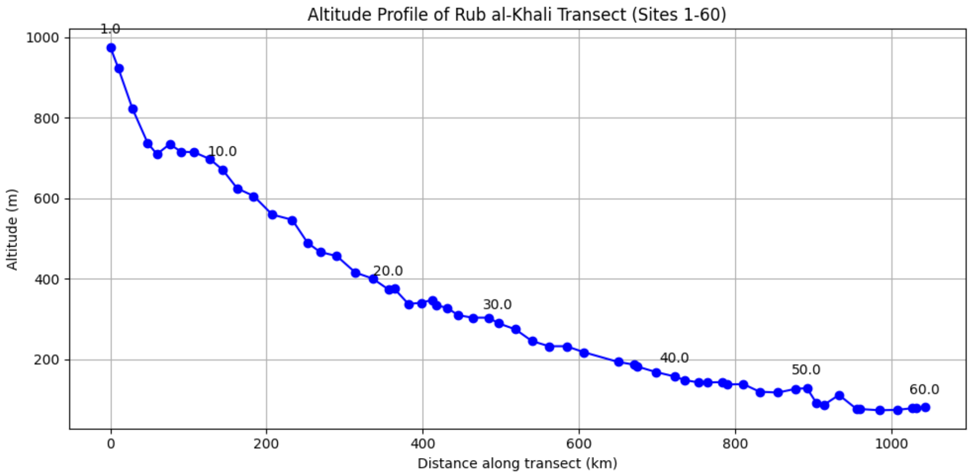

Fig. 1 Altitude profile along the sampling transect across the Rub' al-Khali, from the western escarpment to the eastern border with Oman.

source provenance but do not count them as four additional locations. Because these locations were not revisited on later trips, we exclude their records from analyses on the repeated campaign of sites 1–60. Deterministic corrections to the sample metadata are likewise recorded in a ledger without overwriting the source: in the records `F46Dr2`, `S46Dr2` and `V46Dr2`, the WGS library kit value `NEBNext` had shifted into the site column, and the ledger restores site 46 while retaining the original cell value.

To capture temporal variability, we conducted campaigns across different seasons: late winter 2023 (Trip 1, 64 sites visited), summer 2023 (Trip 2, truncated to 8 sites due to extreme temperatures), winter 2024 (Trip 3, 65 sites visited), summer 2024 (Trip 4, 60 sites visited), and autumn 2025 (Trip 5, 62 sites visited). We recorded temperature, atmospheric pressure, and relative humidity at each visited site in each campaign. We summarize the observational coverage in Table 1. We acquired X-ray fluorescence (XRF) data through two separate workflows that are not interchangeable replicates: handheld in-situ measurements during Trip 5, and laboratory measurements of archived material from all five campaigns. We compiled an environmental table from the original field sheets, recording every correction individually rather than editing the sheets, and retained all 274 non-blank records. The ledger records, for example, that the eight Trip 2 values 34.5–41.9 stood in the humidity column of the source sheet while the temperature column was empty; the field note that the trip ended because of extreme temperature supports reading them as temperatures, and we do not infer Trip 2 pressure or humidity. It also records that the Trip 3 site 21 humidity of 31.321% reproduces the fraction 0.31321 in the original workbook, and that the Trip 5 site 40 source humidity of 194% lies outside the physical range, so its curated value remains missing and the raw value is preserved only in the ledger.

At each site, we collected samples from three distinct soil compartments: bulk surface soil (0 cm depth), bulk subsurface soil (5–10 cm depth, hereafter "deep"), and

Table 1 Coverage of the field environmental logs. Named and auxiliary records are not added to the numeric site catalogue as if they were necessarily distinct physical locations.

Campaign	Season	Field-log records	Measurements available
Trip 1	Late Winter 2023	1–60; 4 named; 15 auxiliary*	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

*The 15 records are appended to the legacy Trip 3 worksheet but are dated March 2023 in the original workbook and are curated as Trip 1 auxiliary/revisit observations.

subsurface soil collected near roots (5–10 cm depth, hereafter "root-adjacent"). The companion ecology analysis [43] calls the same 5–10 cm bulk compartment "shallow-subsurface"; the class label "Deep Soil Sample" denotes that compartment and does not imply a greater sampling depth. We obtained three co-located field replicates per compartment by placing soil into sterile tubes, either by scooping soil directly into the tube or using sterile plastic spoons; these replicates are not independent sites. We preserved samples at −20°C in the field using an Engel MR040F portable freezer. At the end of each campaign, we transferred samples to an −80°C freezer in the laboratory.

We included positive and negative controls for each sampling campaign to mitigate and control for contamination artifacts. One extraction blank was processed with each extraction-day batch, using the same extraction procedure as for the biological specimens, to control for contamination introduced during nucleic acid extraction. For Trips 4 and 5, extraction days were trip-specific (samples from more than one campaign were never extracted together on the same day, though multiple sites from the same campaign could be), so each of their extraction blanks is associated with a single campaign, its extraction date, and its batch; for Trips 1–3, some extractions were repeated and may have combined samples from different trips on one day. Where a corresponding library was successfully generated, the extraction blank was subjected to 16S rRNA gene amplification, indexing, and sequencing.

Marwa/Rumi: confirm the Trips 1 3 extraction days.

The complete ASV table contained 18 sequenced Trip 5 extraction-blank profiles (EB1–EB18), of which 17 were linked by extraction day to 217 biological profiles processed in the same batches, together with six Trip 5 no-template controls (Negative1, Negative2, Negative4–Negative7). The six Trip 4 extraction blanks (SEB, EB1–EB5; one per extraction day, covering 95 sequenced biological samples; one further sample, S40PRr1, failed amplification in three attempts and was not sequenced) were sequenced on a separate run shared with other projects but were not included in the denoising run that produced the canonical table; they were denoised separately with the same pipeline on 30 August 2026 and are distributed with the reproducibility repository. For Trip 4, three of the trip's nine extraction batches were not sequenced, leaving 28 of Trip 4's 60 sites (one of them partially) without a sequenced extraction blank for the corresponding samples.

Two types of PCR-stage negative controls were used to evaluate contamination introduced during amplification and library preparation. No-template controls ("negative controls") consisted of PCR reagents with molecular-grade water substituted for extracted template DNA, evaluating contamination from reagents, water, and the amplification process. PCR blanks consisted of PCR reagents alone, without added water or template. One PCR blank was generated per amplification plate. Across trips, most PCR blanks showed low or undetectable concentration on TapeStation quality control, with profiles dominated by adapter dimer, and were accordingly excluded from sequencing. In Trip 5, none of the PCR blanks were carried through to sequencing, attributed to evaporation of the material during handling, but six no-template controls were sequenced. In Trip 4, one PCR blank and one no-template control were generated but neither was sequenced: the run was shared with samples from Trips 1–3, their index pairs conflicted with biological samples, and the samples were retained. Trip 3 included one sequenced PCR blank (PCRCtrl) and Trips 1 and 2 one each (PCR-Ctrl-Trip1 and the no-template control NTC-2).

A second, field-based negative control consisted of sterile soil, extracted and amplified before the campaign to confirm the absence of PCR product, then taken to the field with the biological specimens. This soil negative control was paired with the trip's positive-control standard as a campaign-control set. Trip 4 did not include this control; for Trip 5 it was sequenced by shotgun metagenomics only. The sequenced controls of Trips 1–3 (Trip 1: one extraction blank, one PCR blank and three labelled standards; Trip 2: two negative controls, one no-template control and two standards; Trip 3: two field negatives, one negative control, one PCR blank and two standards) were denoised with the biological libraries in the July 2025 run and are recorded in the control registry with their labelled roles; the canonical feature table excludes them and retains only the Trip 5 controls. A re-denoising of the Trip 1 and 2 controls on 30 August 2026 showed that Ctrl-1-Trip1, labelled negative, is dominated by the ZymoBIOMICS genera (99.99 % of reads, 24 ASVs), whereas Ctrl-2 and Ctrl-3, labelled positive, and Extraction-Ctrl-Pro-Trip1, labelled extraction blank, carry soil-like profiles; their identities are under review with the laboratory and none of them was used to train the contaminant screen.

Positive-control material and its handling differed among campaigns. Positive controls were generally carried into the field alongside the biological specimens, to validate library amplification and sequencing performance, assess recovery of the expected taxa, and identify whether taxa outside the standard's known composition appeared within the standard's own sequencing profile; since that composition is fully known, any such unexpected taxa would indicate contamination introduced during storage, the campaign, or laboratory processing. Trips 1 and 2 used the ZymoBIOMICS HMW DNA Standard (D6322), a purified-DNA standard added directly at the PCR step rather than passed through extraction. Trip 3 used the whole-cell ZymoBIOMICS Microbial Community Standard (D6300). Trip 4 was not accompanied by a positive standard or a soil-based negative control in the field, and no mock community was taken on this campaign. Laboratory-based controls for Trip 4 consisted of extraction blanks (one per extraction day), a water-based no-template control, a reagent-only PCR blank, and a positive control used to confirm amplification success, checked by gel electrophoresis after first-round PCR and by TapeStation after the second-round (indexing) PCR; only the extraction blanks were sequenced. Trip 5 used the D6300 standard, and a sterile-soil negative control was also taken into the field, but both were processed by shotgun metagenomics rather than 16S. Trips 4 and 5 therefore both lack 16S-specific positive-control validation, for different reasons: Trip 4 had no field-based standard at all (though a PCR-stage-only control was run), and Trip 5's field-based standard was captured only in the shotgun data.

The purified-DNA D6322 standard could be used to assess downstream amplification, library preparation, and sequencing, and, because it was taken into the field, potential contamination from field handling and storage. Because it was introduced directly into PCR, it could not assess cell lysis or extraction recovery. The whole-cell D6300 standard could additionally assess extraction efficiency and differential cell lysis. D6322 contains seven bacterial genomes at 14 % genomic DNA each and *Saccharomyces cerevisiae* at 2 %; D6300 contains eight bacterial genomes at 12 % each and two yeasts at 2 % each. These manufacturer genomic-DNA fractions are composition specifications, not expected 16S read fractions. Positive standards were not used as ecological samples, nor to train the extraction-blank contaminant screen. Seven positive control samples from Trips 1–3 were used to evaluate genus-level recovery.

Author-confirmed controls are recorded in a ground-truth table; identity, batch-membership and assay-applicability claims that await laboratory confirmation are represented in the graph as evidence-bearing dispositions rather than as ground truth, and no field-blank instance is asserted because no sterile-bag inventory was recovered.

2.2 DNA extraction and sequencing

Genomic DNA was extracted from 0.9–1.2 g of soil using either the QIAGEN PowerLyzer PowerSoil Kit (catalogue no. 12855-50) or the QIAGEN DNeasy PowerSoil Pro Kit (catalogue no. 47014). The PowerLyzer PowerSoil Kit was used for the initial extraction of samples from Trips 1 and 2. From Trip 3 onward, samples were extracted using the DNeasy PowerSoil Pro Kit. The QIAcube Connect was introduced for routine extraction starting with Trip 4; samples from Trips 1–3 were also processed using the QIAcube Connect when re-extraction was required. The extraction method and re-extraction history were recorded for each specimen when available.

Frozen soil samples were thawed on ice and processed immediately after thawing. Before homogenization, samples were incubated at 65°C for 10 min with agitation at 1,200 rpm, then homogenized using a PowerLyzer 24 Homogenizer according to the manufacturer's protocol. The remaining purification steps were performed either manually or using a QIAcube Connect instrument, following the manufacturer's protocol. DNA was eluted in 50 µL of the elution buffer supplied with the respective kit.

One extraction blank was processed alongside each extraction-day batch and carried through the same extraction procedure as the biological samples. Because extraction days were trip-specific, each extraction blank is associated with a single campaign. Extraction blanks were quantified and included in the downstream 16S rRNA gene amplification and sequencing workflow when a corresponding library was generated; exceptions are described above.

Extracted DNA was quantified using the Qubit dsDNA High Sensitivity assay, with fluorescence measured using a Varioskan LUX multimode plate reader (Thermo Fisher Scientific). A numerical concentration was recorded for every extract. Progression to indexing and sequencing was determined by the presence of an amplicon of the expected size after first-round PCR, rather than by a minimum extracted DNA concentration threshold. DNA concentrations and processing information were recorded in a central project spreadsheet. Extracted DNA was stored at −20°C until PCR amplification.

The V3–V4 region of the bacterial 16S rRNA gene was amplified using the Bakt_341F (`CCTACGGGNGGCWGCAG`) and Bakt_785R (`GACTACHVGGGTATCTAATCC`) primers [44], both carrying Illumina overhang adapters. First-round PCR was performed using TaKaRa Ex Taq DNA Polymerase (catalogue no. RR001B). Each reaction had a total volume of 50 µL and contained 5 µL of extracted DNA and 2 µL of each 10 µM primer stock, giving a final concentration of 0.4 µM per primer. Cycling conditions were: initial denaturation at 95°C for 3 min; 30 cycles of 95°C for 30 s, 63°C for 30 s and 72°C for 1 min; and a final extension at 72°C for 6 min.

First-round PCR products were assessed by electrophoresis on a 1% agarose gel. Samples showing an amplicon at the expected size of approximately 500 bp were classified as passing and were then indexed. Samples without a visible product were classified as failed; these were first re-amplified with an increased DNA input, and samples that still failed were re-extracted at the end of the workflow.

Dual indices and Illumina sequencing adapters were added in a second-round PCR using the QIAGEN Multiplex PCR Kit (catalogue no. 206143), following the Illumina 16S Metagenomic Sequencing Library Preparation protocol [45]: initial activation at 94°C for 15 min; 8 cycles of 94°C for 30 s, 55°C for 30 s and 72°C for 30 s; final extension at 72°C for 5 min.

Indexed libraries were purified using Agencourt AMPure XP beads, quantified using the Qubit dsDNA High Sensitivity assay, and assessed for fragment size using an Agilent 4200 TapeStation with the D5000 ScreenTape assay. Libraries were pooled by concentration and fragment size. Amplicon pools were sequenced on an Illumina NovaSeq 6000 using an SP flow cell with a 500-cycle reagent kit to generate 2×250 bp paired-end reads. Sample processing and library preparation were performed in the PI's laboratory and sequencing was conducted at the Bioscience Core Lab at KAUST.

2.3 Sequencing data processing

We processed raw amplicon reads with nf-core/ampliseq (v2.14.0) [46, 47], applying the DADA2 algorithm [48] for error correction and denoising, and assigning taxonomy against the SILVA database (v138.2) [49]. The dataset comprises 1,268 sequencing libraries from 1,238 samples; 30 samples were sequenced twice. This resulted in a canonical feature table of 351,472 amplicon sequence variants (ASVs) prior to contaminant removal, covering samples from Trips 1–5 and 24 control samples. Of its 1,271 profiles, 1,247 are biological; quality filtering excludes the 24 control profiles and 10 biological profiles with fewer than 1,000 reads, leaving the 1,237 profiles used by the companion ecology analysis. The complete table is distributed unfiltered.

We applied an assay-aware contamination audit following recommendations for low biomass studies [50]. We pooled 17 extraction day blanks that were processed on the same day as Trip 5 samples, and tested each ASV for enrichment in extraction blanks using a one-sided Fisher's exact test ($p < 0.1$, uncorrected), requiring presence in at least two blanks. Because each extraction batch is represented by a single blank, within-batch prevalence could not be estimated. This resulted in a candidate set of 351 ASVs, applied only to the 217 biological samples linked to those batches. We used seven positive control samples from Trips 1–3 to evaluate genus-level recovery, but not to train the contaminant classifier. The primary table remains unfiltered, and the candidate set is used only in a bounded sensitivity analysis (Section 6).

We calculated genome-level functional potential with PICRUSt2 (v2.4.1) [51] and used it for the companion analysis. The execution manifest with the exact commands, hidden-state method, edge exponent, seed and thresholds, is archived with the release. The predictions came from two separately executed input sets, the Trips 1–4 ampliseq output and the corrected Trip 5 ampliseq output (330,830 ASVs); they were not generated from the later 351,472-ASV combined canonical table.

2.4 Taxonomy normalization and mapping

Taxonomic labels are text strings, and the same string can denote different taxa across databases and releases, and one taxon can be referred to by several strings [52]. We therefore map NCBI Taxonomy [42] identifiers to SILVA lineages alongside every assignment. Taxonomic assignments come from SILVA v138.2 (Section 2), and we treat the original classifier lineage and ASV identifier as the primary evidence. These assignments are seven rank fields (domain–species), and some are followed by a separate species assignment field and a numeric confidence. We assert the species from the ranked lineage and retain the additional species call in a provenance ledger. We normalize the seven ordered rank fields by removing rank prefixes and representing empty and unclassified entries explicitly.

A fail-closed normalization step evaluates the historical SILVA-to-NCBI candidates against a pinned NCBI Taxonomy release (the OBO Foundry NCBITaxon release `2025-03-13`, built from the NCBI organismal taxonomy of 2025-03-01). An identifier is retained only when it resolves, has a compatible rank and label or documented synonym, and its ancestor chain is compatible with all retained higher-rank identifiers in that source lineage. A missing, ambiguous or incompatible candidate is replaced by a deterministic project class scoped to the complete truncated source lineage and rank, rather than by the nearest ancestor's identifier, which would conflate a named child with its broader ancestor. Each decision records the source lineage and rank, candidate identifier, validation outcome, final identifier and reason. The output mapping covers every lineage represented in both the taxonomy export and the feature table; otherwise generation stops.

Existing project identifiers are retained only when all explicit historical uses share one normalized rank, label and nearest explicit parent context; a reused identifier that acquires more than one corrected context is replaced in every affected use by contextual classes. Conflicting historical evidence is therefore contextualized rather than resolved according to file order. The mapping builder emits corrected mappings, a decision ledger, a clean project taxonomy module and a manifest containing input and output checksums, and the abundance-ABox generator accepts only a passing, checksum-matched manifest.

We evaluated cross-resource alignment to GTDB [53] and iNaturalist separately from this SILVA-to-NCBI layer, in an earlier exploration whose candidate tables are not part of the release. iNaturalist enters this evaluation because the plant specimens sampled alongside the rhizosphere profiles were identified with it, so its taxon labels supply the host-plant context for those samples rather than an alternative prokaryotic taxonomy. The conclusion that exploration supports is a negative one: homonyms, *Candidatus* names, placeholder taxa, GTDB split taxa, rank conflicts and database version changes can all produce identical labels for non-identical concepts, so label agreement alone does not establish extensional identity. The canonical release modules therefore contain no `owl:equivalentClass` assertions generated from those lexical candidates. Unreviewed GTDB and iNaturalist candidates are excluded from the asserted crosswalk, and we claim no mapping precision or recall because a stratified manual gold standard has not been completed. This conservative representation permits queries over the original SILVA lineage and validated NCBI classes without turning lexical similarity, a shared label or a supplementary species call into unsupported taxonomic identity.

2.5 Shotgun-derived companion inputs

Two of the tabular companion-analysis inputs distributed with the deposit (Section 4) derive from a separate shotgun metagenomic study of the same campaigns rather than from the 16S workflow above. A catalogue of metagenome-assembled genomes was assembled, binned and dereplicated in that study. The shotgun data raise their own assembly, binning and profiling challenges and will be reported later as a separate study; its assembly and binning parameters and its read accessions belong to that report and are not recorded in this deposit. From that catalogue we distribute (i) 150 per-library CoverM [54] genome-relative-abundance tables (`metadata/metagenome/coverm_profiles.tar.gz`; one column per shotgun library, in percent of reads, with the unmapped fraction retained as its own row) and (ii) the eggNOG-mapper v2.1.12 annotation table [55] (`metadata/metagenome/eq.emapper.annotations.gz`) of the concatenated predicted proteome, run on the KAUST Ibex cluster on 14 June 2026, from which a 990-genome by KEGG Orthology copy-number matrix is built for the genomes present in both inputs. These files are provided as records for downstream re-execution of the ecology paper's encoded-function test, not as graph modules, and the deposit does not by itself make the upstream assembly, binning or mapping reproducible.

2.6 Environmental and climate data

We retrieved historical climate data from the Open-Meteo archive endpoint [38] for each sampling site using its geographic coordinates, yielding 12,936 monthly records encompassing temperature, total precipitation, rainfall and relative humidity, as well as annual records. We ran two separately configured acquisitions: the monthly product covers 1 January 2022 to 20 January 2026 and includes mean relative humidity, whereas the daily product covers 1 January 2022 to 1 February 2026 and reports temperature, total precipitation and rainfall, but not humidity. Furthermore, field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO) [41].

2.7 XRF acquisition, processing and reconciliation

We acquired XRF data through two complementary workflows. During Trip 5, field researchers used a Vanta handheld instrument to perform in-situ field measurements. The source log contains 106 entries across 59 sites; 71 complete sessions at 58 sites were retained, and nine of those sites have repeated complete sessions. The field measurements do not refer to a specific specimen, so field and laboratory linkage is site-level rather than a same aliquot comparison. Archived material from all five campaigns was subsequently measured in the laboratory. The laboratory inputs contain 547 selected sample records from Trips 1–4 and 178 from Trip 5 (725 records in total), and all 725 are used in the analyses; the retired 158- and 705-row artifacts (an earlier Trip 5 table and the cross-trip table built from it) are not analytical inputs.

Both laboratory sources occasionally report the same formula more than once for one sample: the Trips 1–4 workbooks list a formula under several reporting sections (for example the element and oxide sections), and the Trip 5 workbook contains one sample (V14Dr1) with two readings. The two source-specific parsers resolve these repeats differently, taking the largest positive value for Trips 1–4 and the last reported row for Trip 5; neither policy is instrument-prescribed. The provenance audit reproduced all 13,390 reported Trips 1–4 cells and all 4,293 reported Trip 5 cells under these rules. In practice the two rules give the same value everywhere except three analyte cells (Al, Na and Si) of that one Trip 5 record, because repeated Trips 1–4 rows always carry identical values; a single rule would therefore change one of the 725 laboratory records, and the release discloses both policies rather than implying a common calibrated procedure.

All XRF instrument channels (e.g. element, oxide) were mapped to a chemical identifier in the pinned ChEBI [40] release 247 (3 December 2025) and cross-checked against a PubChem [56] property snapshot retrieved on 23 July 2026. An executable audit rejects identifiers that cannot be found, that have been deprecated, or whose formula does not match the channel. For a descriptive cross-workflow diagnostic, we took the median of repeated field sessions at each site and compared it with the Trip 5 Deep laboratory record at the 58 sites measured by both workflows. Rank agreement was modest: for the 12 analytes reported positive by both workflows at more than 10 of the 58 matched sites, the Spearman correlation between field and laboratory values had a median of 0.21 (range −0.40 to 0.48; `evidence/xrf_audit/xrf_field_lab_agreement.tsv`). We do not infer calibration between the two workflows, and no analysis treats field and laboratory values as being on the same scale.

2.8 Archived-soil pH acquisition and admission

We measured archived dry soil after sieving below 2 mm. Each assay used an independent 1 g scoop suspended in 2.5 mL 0.01 M $CaCl_2$. Electrodes were calibrated at two points (pH 7 and 10), and we recorded the electrode slope and pH-10 buffer readback for every session. A measurement was admitted only if its session was calibrated, the electrode slope was between 95% and 102%, and the read-back was within 0.1 pH units of nominal.

This resulted in 1,168 recordings of which 712 measurements were admitted. The remainder consisted of 356 rows that lacked measurement, 45 marked as depleted, 36 with ambiguous dates, and 19 numeric rows with incomplete quality control records. We neither inferred missing quality-control fields nor imputed unavailable values. The workbook has no recorded session identifier. We therefore reconstructed 29 sessions from the unique combinations of trip, accepted measurement date, electrode slope, and pH-10 read-back, as these groupings preserve the quality-control context available in the source, but are inferred rather than recorded laboratory batch identifiers.

2.9 Schema validation

To ensure structural integrity within the RDF graph, we used Shape Expressions (ShEx) [57]. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived-soil pH processes, values and sessions, and taxonomy mappings.

We integrated this schema validation into our deployment workflow using a Shape Map. The Shape Map functions as a binding directive, explicitly linking node selectors (e.g., all instances of the sampling site class) to their respective ShEx shape definitions. The workflow tests each newly generated segment of instance data against the schema. It terminates the deployment if it detects missing mandatory properties (e.g., GPS coordinates for a sampling site) or invalid datatypes. This constraint application guarantees that all generated data conforms to the expected structural properties. We implemented this validation using the Apache Jena ShEx library (v4.10.0) invoked from custom Groovy scripts. The multi-gigabyte taxonomy ABox is excluded from this memory-bounded ShEx run; the exact generated file is instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process–dataset–quality–value relationship.

2.10 Logical consistency (OWL reasoning)

We checked logical consistency using the ELK reasoner [58]. The ELK reasoner operates on the OWL 2 EL profile and detects unsatisfiable classes and contradictory axioms within ontologies. ELK implements a polynomial-time algorithm, facilitating the consistency validation of large knowledge bases. However, this performance benefit necessitates a tradeoff in logical expressivity, as the OWL 2 EL profile does not support complex constraints such as universal quantification or inverse object properties found in ontologies like the Semanticscience Integrated Ontology (SIO). We employed ELK instead of more expressive reasoners (e.g., HermiT [59]) because verifying the large knowledge graph we created with a more expressive reasoner proved impossible due to computational complexity. We invoked ELK through the OWL API (v4.5.26) with the ELK OWL API module (v0.4.3) from a Groovy script, loading the tractable set of OWL modules into a single merged ontology. The current run is memory-bounded and excludes the multi-gigabyte taxonomy ABox and large reference modules, so a successful run establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. This approach enabled rapid continuous integration during ontology development.

2.11 Reproducible workflow

Release generation, validation and manuscript-derived statistics are orchestrated from one Nextflow entry point (`main.nf`), with separate local and cluster profiles that change execution infrastructure without changing process definitions. The workflow regenerates the site, environmental-measurement, sample, XRF, DNA, sequence-record, quality-control, laboratory-control and pH modules from their declared sources and requires every byte stream to match the staged release candidate, so the deposited modules and the manuscript statistics are produced from the same auditable run.

3 Knowledge representation

We developed a formal knowledge base to integrate the different types of data generated in this project. The semantic layer encodes the full provenance chain from physical sampling to digital data products, enabling cross-domain queries and ensuring data interoperability through shared vocabularies.

3.1 Architecture

The knowledge base is constructed in Description Logic [60] using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox defines the ontology of the knowledge graph using OWL 2 DL [36], while the ABox contains the instance data generated from the campaign. We deployed the complete knowledge base across three named RDF graphs: an ontology graph (17,025 triples) containing the TBox, a knowledge base graph containing the asserted instance data, and a taxonomy reference graph containing the aligned taxonomy module.

3.2 Prefixes used in listings

For brevity, the Turtle and SPARQL listings throughout the paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against the public endpoint. With these declarations in place, identifiers such as `https://rubalkhali.science/kb/RAK_P100001` are abbreviated to `rak:P100001` and `http://semanticscience.org/resource/SIO_000291` to `sio:000291` without loss of information.

3.3 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO) [39] as the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific

Table 2 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
`rak:`	`https://rubalkhali.science/kb/RAK_`
`sio:`	`http://semanticscience.org/resource/SIO_`
`pato:`	`http://purl.obolibrary.org/obo/PATO_`
`uo:`	`http://purl.obolibrary.org/obo/UO_`
`envo:`	`http://purl.obolibrary.org/obo/ENVO_`
`ncbitaxon:`	`http://purl.obolibrary.org/obo/NCBITaxon_`
`obo:`	`http://purl.obolibrary.org/obo/`
`rdf:`	`http://www.w3.org/1999/02/22-rdf-syntax-ns#`
`rdfs:`	`http://www.w3.org/2000/01/rdf-schema#`
`xsd:`	`http://www.w3.org/2001/XMLSchema#`
`geo:`	`http://www.opengis.net/ont/geosparql#`
`dcterms:`	`http://purl.org/dc/terms/`

domains, ensuring semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (see Section 8).

The canonical terminology module declares 333 classes: 297 local project classes and 36 referenced classes (32 SIO and four PATO [61]). It also declares 20 object properties (11 project-local and nine SIO) and 35 datatype properties (34 project-local and one SIO). These are declaration counts from the staged canonical module, not counts of the imported ontologies or of whole-graph axioms. This compact semantic structure maintains precise alignment with our standardized schemas while minimizing redundant inferences.

3.3.1 Measurement pattern

We model a quantitative measurement using SIO's ontology design pattern for measurement processes: a measurement is a process that generates a value for a specific quality of a target entity.

$$\text{MeasurementProcess} \sqsubseteq \text{Process} \tag{1}$$

$$\text{Quality} \sqsubseteq \text{Attribute} \tag{2}$$

$$\text{Quantity} \sqsubseteq \text{InformationContentEntity} \tag{3}$$

$$\text{MeasurementProcess} \sqsubseteq \exists \text{hasTarget.Entity} \sqcap \exists \text{hasOutput.Quantity} \tag{4}$$

$$\text{Quantity} \sqsubseteq \exists \text{hasValue.xsd:double} \sqcap \exists \text{hasUnit.Unit} \tag{5}$$

We identified all units with classes from the Units Ontology (UO) [62]. Qualities are expressed using the Phenotype and Trait Ontology (PATO) [61] and linked to SIO as subclasses of the *Quality* class. To illustrate this pattern, we provide an example represented in Turtle format. The example (Listing 1) demonstrates an environmental temperature measurement taken at a sampling site.

Listing 1 Turtle representation of an environmental temperature measurement.

```
rak:P000001 a rak:0000006 ; # Measurement Process
  sio:000068 rak:3000001 ; # isPartOf (site visit)
  sio:000229 rak:4000001 . # hasOutput (Value)

rak:3000001 a rak:0000003 ; # Site Visit
  sio:000291 rak:1000001 . # hasTarget (Sampling Site)

rak:4000001 a rak:0000010 ; # Quantity
  sio:000221 uo:0000027 ; # hasUnit (degree Celsius)
  rak:2000003 "20.7"^^xsd:double ; # hasValue
  sio:000215 rak:5000001 . # isMeasurementValueOf (Quality)

rak:5000001 a pato:0000146 ; # Temperature Quality
  sio:000011 rak:1000001 ; # isAttributeOf
  sio:000216 rak:4000001 . # hasMeasurementValue
```

In words, an instance `rak:P000001` of a measurement process is part of the site visit `rak:3000001` that targets the sampling site `rak:1000001`, and produces as output the quantity `rak:4000001`. That quantity carries the literal value `20.7`, is reported in degrees Celsius (`uo:UO_0000027`), and refers to a temperature quality `rak:5000001` which is in turn an attribute of the same site. In SIO, `sio:000215` is the value-to-quality relation and `sio:000216` is its quality-to-value inverse. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies.

3.3.2 Sampling pattern

A sampling event is a process that derives a sample from a feature of interest (the site).

$$\text{SamplingProcess} \sqsubseteq \text{Process} \tag{6}$$

$$\text{Sample} \sqsubseteq \text{MaterialEntity} \tag{7}$$

$$\text{SamplingProcess} \sqsubseteq \exists \text{hasTarget.Site} \sqcap \exists \text{hasOutput.Sample} \tag{8}$$

$$\text{Sample} \sqsubseteq \exists \text{isDerivedFrom.Site} \tag{9}$$

Listing 2 illustrates the semantic representation of a surface soil sample collected during Trip 1.

Listing 2 Turtle representation of a sampling process.

```
rak:P100001 a rak:0000024 ; # Sampling Process
  sio:000291 rak:1000001 ; # hasTarget
  sio:000229 rak:7000001 . # hasOutput (Collection)

rak:6000001 a rak:0000021 ; # Deep Soil Sample
  sio:000244 rak:1000001 . # isDerivedFrom
```

Read in prose, an instance `rak:P100001` of a sampling process targets the site `rak:1000001` and yields a collection `rak:7000001` as its output; the deep soil sample rak:6000001 is in turn asserted to be derived from the same site. The redundancy between hasTarget on the process and isDerivedFrom on the sample is intentional: the former records the action, the latter records the resulting provenance edge directly on the material entity, so queries can traverse from sample to site without joining through the sampling event.

3.3.3 Experimental protocol pattern

We model experimental protocols as processes specified by a protocol entity. Each process (e.g., DNA extraction, library preparation) consumes inputs and produces outputs, explicitly linking agents and devices. The pattern instantiates SIO's general *Process–hasInput/hasOutput–Object* idiom and adds a specification link to a reusable protocol description.

$$\text{ProtocolExecution} \sqsubseteq \text{Process} \tag{10}$$

$$\text{Protocol} \sqsubseteq \text{InformationContentEntity} \tag{11}$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{isSpecifiedBy.Protocol} \tag{12}$$

$$\text{ProtocolExecution} \sqsubseteq \exists \text{hasInput.MaterialEntity} \sqcap \exists \text{hasOutput.MaterialEntity} \tag{13}$$

Listing 3 demonstrates the formalization of a DNA extraction process.

Listing 3 Turtle representation of a DNA extraction protocol.

```
rak:P200001 a sio:000994 ; # Scientific Protocol Execution
  sio:000339 rak:DNA_ext_powersoil ; # isSpecifiedBy
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:6800001 a rak:0000040 ; # DNA Extract
  sio:000008 rak:4900001 . # hasAttribute (Concentration Quality)
```

In words, the protocol-execution instance rak:P200001 is specified by the named extraction protocol rak:DNA_ext_powersoil, takes the soil sample rak:6000001 as input, and produces the DNA extract rak:6800001 as output; the resulting extract carries a concentration quality (rak:4900001) which is itself quantified through a separate measurement process following the pattern of Equation 1.

3.3.4 Taxonomic abundance pattern

Taxonomic profiles represent collections of taxon observations containing associated relative and absolute abundances. Instead of modeling the feature table as a flat matrix, we formalize the entire table as a bioinformatic workflow that produces a dataset containing individual taxonomic observations. Here, we use the merged taxonomy (see Section 2) to assign specific NCBI or GTDB classes as the target of each observation. Listing 4 shows how we map an entire sample's feature table row into the knowledge graph.

For instance, the sequence analysis for a specific sample yields an absolute sequence count associated with a taxonomic class from the NCBI Taxonomy. We construct an abundance quality that targets the explicit taxonomic identifier, binding the observational count back to the biological workflow. Formally, the pattern instantiates SIO's *Collection–hasMember–Thing* and *Quantity–isMeasurementValueOf–Quality* idioms:

$$\text{BioinformaticWorkflow} \sqsubseteq \text{Process} \tag{14}$$

$$\text{TaxonomicDataset} \sqsubseteq \text{Collection} \tag{15}$$

$$\text{BioinformaticWorkflow} \sqsubseteq \exists\text{hasInput.SequenceDataset} \sqcap \exists\text{hasOutput.TaxonomicDataset} \tag{16}$$

$$\text{TaxonomicDataset} \sqsubseteq \forall\text{hasMember.AbundanceObservation} \tag{17}$$

$$\text{AbundanceObservation} \sqsubseteq \text{Quantity} \sqcap \exists\text{isMeasurementValueOf.AbundanceQuality} \tag{18}$$

$$\text{AbundanceQuality} \sqsubseteq \exists\text{isAttributeOf.Taxon} \tag{19}$$

Listing 4 Turtle representation of taxonomic abundance formalized from a feature table.

```
rak:P290001 a rak:0000071 ; # Bioinformatic Workflow
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:7290001 . # hasOutput (Taxonomic Dataset)

rak:7290001 a rak:0000074 ; # Taxonomic Dataset
  sio:000059 rak:4400001 . # hasMember (Observation Value)

rak:4400001 a rak:0000076 ; # Absolute Abundance Measurement Value
  sio:000215 rak:5400001 ; # isMeasurementValueOf
  rak:2000026 "15"^^xsd:integer . # hasAbsoluteAbundanceValue

rak:5400001 a rak:0000078 ; # Absolute Abundance Quality
  sio:000011 ncbitaxon:286 . # isAttributeOf (Pseudomonas)
```

Verbalized, the bioinformatic-workflow instance `rak:P290001` consumes a FASTQ dataset (`rak:7900001`) and produces the taxonomic dataset `rak:7290001`; that dataset has as one of its members the quantity `rak:4400001`, which records an absolute read count of `15` and refers to an absolute abundance quality that is in turn an attribute of *Pseudomonas* (`obo:NCBITaxon_286`). One row of a feature table is therefore expanded into a four-node sub-graph that preserves both the classifier output and its provenance.

3.3.5 XRF geochemistry pattern

We model XRF analysis as a process that targets a soil sample and produces multiple measurement values, each corresponding to the concentration of a specific chemical analyte. We map each analyte to the Chemical Entities of Biological Interest (ChEBI) ontology [40] as well as to the PubChem database [56] (Section 2.7). Out of 93 monitored geochemical parameters, we mapped 80 analytes directly to corresponding ChEBI classes and aligned 92 parameters to PubChem Compound Identifiers. The XRF pattern is a specialization of the measurement pattern (Equation 1), with the target restricted to a soil sample and the produced quantity bound to an analyte-specific quality:

$$\text{XRFMeasurementProcess} \sqsubseteq \text{MeasurementProcess} \tag{20}$$

$$\text{XRFMeasurementProcess} \sqsubseteq \exists \text{hasTarget.SoilSample} \sqcap \exists \text{hasOutput.ConcentrationQuantity} \tag{21}$$

$$\text{ConcentrationQuantity} \sqsubseteq \text{Quantity} \sqcap \exists \text{isMeasurementValueOf.AnalyteQuality} \tag{22}$$

$$\text{AnalyteQuality} \sqsubseteq \exists \text{isAttributeOf.SoilSample} \tag{23}$$

Listing 5 illustrates an XRF measurement for a specific analyte.

Listing 5 Turtle representation of an XRF geochemistry measurement.

```
rak:P300001 a rak:0000025 ; # XRF Measuring Process (laboratory)
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:4300001 . # hasOutput

rak:4300001 a rak:0000567 ; # CaO Concentration Measurement Value
  sio:000215 rak:5300001 ; # isMeasurementValueOf
  rak:2000012 "58.2"^^xsd:double . # hasConcentrationValue (no unit)

rak:5300001 a rak:0000167 ; # CaO Concentration Quality
  sio:000011 rak:6000001 . # isAttributeOf
```

In words, the XRF measurement process `rak:P300001` takes the soil sample `rak:6000001` as its input and produces the quantity `rak:4300001`, which records a concentration value of `58.2` (the instrument exports document no unit, so none is asserted) and refers to an analyte-specific quality `rak:5300001`. Because the analyte quality is typed with a dedicated subclass per element, the same pattern instance can be reused unchanged across all 91 geochemical analytes by varying only the quality class IRI.

3.3.6 Sampling site and biome assignment pattern

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. We use both an existential and a universal restriction so that the asserted biome both holds and is the only admissible value, which preserves the closed-world intent of the field record.

$$\text{SamplingSite} \sqsubseteq \text{Site} \tag{24}$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasBiome.ENVO:Biome} \sqcap \forall \text{hasBiome.ENVO:Biome} \tag{25}$$

$$\text{SamplingSite} \sqsubseteq \exists \text{hasEnvironmentalFeature.ENVO:EnvironmentalFeature} \tag{26}$$

Listing 6 Turtle representation of a sampling site grounded in ENVO.

```
rak:1000001 a rak:0000002 ; # Sampling Site
  rdfs:label "Site 1" ;
  geo:asWKT
    "POINT(45.166511388888885 19.149770555555556)"^^geo:wktLiteral ;
  a [ a owl:Restriction ;
      owl:onProperty rak:2000001 ; # hasBiome
      owl:someValuesFrom envo:01000179 ], # desert biome
    [ a owl:Restriction ;
      owl:onProperty rak:2000002 ; # hasEnvironmentalFeature
      owl:someValuesFrom envo:01000018 ]. # gravel
```

In words, the sampling site `rak:1000001` carries a label, a WKT geometry, and is asserted both to belong to the ENVO desert biome and to exhibit a gravel environmental feature. The combined existential and universal restriction on `hasBiome` (Equation 24) ensures that any reasoner treats the asserted biome as the only admissible value for that site.

3.3.7 Sequencing workflow pattern

Sequencing data are produced by a chain of laboratory and computational processes that the knowledge graph represents as a sequence of *Process–hasInput/hasOutput–Object* steps: a DNA extraction process consumes a soil sample and produces a DNA extract; a library preparation process consumes the DNA extract and produces an amplicon library; a sequencing process consumes the amplicon library and produces a FASTQ dataset whose members are sequence reads. Each process is specified by a protocol and may carry agents and device participants, reusing the experimental protocol pattern of Equation 10.

$$\text{DNAExtractionProcess} \sqsubseteq \text{ProtocolExecution} \tag{27}$$

$$\text{DNAExtractionProcess} \sqsubseteq \exists\text{hasInput.SoilSample} \sqcap \exists\text{hasOutput.DNAExtract} \tag{28}$$

$$\text{LibraryPreparationProcess} \sqsubseteq \text{ProtocolExecution} \tag{29}$$

$$\text{LibraryPreparationProcess} \sqsubseteq \exists\text{hasInput.DNAExtract} \sqcap \exists\text{hasOutput.AmpliconLibrary} \tag{30}$$

$$\text{SequencingProcess} \sqsubseteq \text{ProtocolExecution} \tag{31}$$

$$\text{SequencingProcess} \sqsubseteq \exists\text{hasInput.AmpliconLibrary} \sqcap \exists\text{hasOutput.FASTQDataset} \tag{32}$$

$$\text{FASTQDataset} \sqsubseteq \text{Dataset} \sqcap \forall\text{hasMember.SequenceRead} \tag{33}$$

Listing 7 Turtle representation of the sequencing workflow from soil sample to FASTQ dataset.

```
rak:P200001 a rak:0000309 ; # DNA Extraction Process
  sio:000339 rak:L000001 ; # isSpecifiedBy (PowerSoil protocol)
  sio:000230 rak:6000001 ; # hasInput (Soil Sample)
  sio:000229 rak:6800001 . # hasOutput (DNA Extract)

rak:P250001 a rak:0000065 ; # Library Preparation Process
  sio:000339 rak:L000010 ; # Illumina 16S Prep Guide
  sio:000230 rak:6800001 ; # hasInput (DNA Extract)
  sio:000229 rak:6900001 . # hasOutput (Amplicon Library)

rak:P280001 a rak:0000066 ; # Sequencing Process
  sio:000230 rak:6900001 ; # hasInput (Amplicon Library)
  sio:000229 rak:7900001 . # hasOutput (FASTQ Dataset)

rak:7900001 a rak:0000063 ; # FASTQ Dataset
  rdfs:seeAlso <https://www.ebi.ac.uk/ena/browser/view/ERR1234567> .
```

Read as a chain, the DNA extraction process `rak:P200001` consumes the soil sample `rak:6000001` and produces the DNA extract `rak:6800001`; that extract is the input of the library preparation `rak:P250001`, which yields the amplicon library `rak:6900001`; finally the sequencing process `rak:P280001` consumes the library and produces the FASTQ dataset `rak:7900001`, which is cross-referenced to its ENA accession via `rdfs:seeAlso`. Each link in the chain is the same `hasInput`/`hasOutput` pattern, so a single SPARQL traversal can recover the full provenance from any node.

3.3.8 Sequencing quality control pattern

Sequencing quality control is represented as an additional computational process that takes a FASTQ dataset as input and produces multiple measurement values, each instantiating the generic measurement pattern of Equation 1. Metrics (sequence count, GC content, duplicate rate, read length) are split into forward (R1) and reverse (R2) subclasses so that the provenance of paired-end runs is preserved.

$$\text{SequencingQCProcess} \sqsubseteq \text{Process} \tag{34}$$

$$\text{SequencingQCProcess} \sqsubseteq \exists\text{hasInput.FASTQDataset} \sqcap \exists\text{hasOutput.QCMeasurementValue} \tag{35}$$

$$\text{QCMeasurementValue} \sqsubseteq \text{Quantity} \sqcap \exists\text{isMeasurementValueOf.QCQuality} \tag{36}$$

$$\text{QCQuality} \sqsubseteq \exists\text{isAttributeOf.FASTQDataset} \tag{37}$$

Listing 8 Turtle representation of a sequencing QC measurement (forward-read GC content).

```
rak:P350001 a rak:0000200 ; # Sequencing QC Process
  sio:000230 rak:7900001 ; # hasInput (FASTQ Dataset)
  sio:000229 rak:4500001 . # hasOutput (QC Value)

rak:4500001 a rak:0000217 ; # Forward GC Content Value
  sio:000215 rak:5500001 ; # isMeasurementValueOf
  sio:000221 uo:0000187 ; # Percent
  rak:2000074 "54.2"^^xsd:double . # hasForwardGCContent

rak:5500001 a rak:0000205 ; # Forward GC Content Quality
  sio:000011 rak:7900001 . # isAttributeOf (FASTQ Dataset)
```

In words, the QC process `rak:P350001` takes the FASTQ dataset `rak:7900001` as input and produces the measurement value `rak:4500001`, which records a GC content of `54.2` percent and refers to a forward-read GC-content quality `rak:5500001` that is in turn an attribute of the same FASTQ dataset. Splitting the quality class into forward and reverse subclasses preserves the read direction, so that paired-end QC metrics can be queried independently per orientation.

3.3.9 Laboratory-control pattern

A laboratory control is an occurrence-specific material entity, not a property of a campaign and not the string used to label a library. Each physical control carries its own instance of a positive, negative, extraction-blank or PCR-blank role, and a laboratory process realizes that role while consuming the material as input and target and producing a derived material or dataset. Processes are grouped into batches, and, because an extraction day can contain samples from several campaigns, neither the control nor its batch is linked directly to a trip.

$$\text{ControlMaterial} \sqsubseteq \text{MaterialEntity} \tag{38}$$

$$\text{ControlRole} \sqsubseteq \text{Role} \tag{39}$$

$$\text{ControlMaterial} \sqsubseteq \exists \text{hasRole.ControlRole} \tag{40}$$

$$\begin{aligned}\text{LaboratoryProcess} \sqsubseteq\ & \exists \text{hasInput.ControlMaterial} \sqcap \exists \text{hasTarget.ControlMaterial} \\ & \sqcap \exists \text{realizes.ControlRole}\end{aligned} \tag{41}$$

$$\text{LaboratoryProcess} \sqsubseteq \exists \text{isPartOf.ProcessingBatch} \tag{42}$$

Listing 9 instantiates the pattern for an extraction blank. The blank material `rak:C000001` bears an extraction-blank role `rak:R000001` through `sio:000228` (*has role*); the extraction process `rak:P400001` takes the blank as input and target, realizes the role through `sio:000355`, and is part of the extraction batch `rak:B000001`. Asserting the role as an individual rather than as a class membership keeps each physical control distinct even when labels are reused across campaigns.

Listing 9 Turtle representation of an extraction-blank control and its role realization.

```
rak:C000001 a rak:0000302 ; # Blank Control Material
  sio:000228 rak:R000001 . # hasRole

rak:R000001 a rak:0000306 ; # Extraction Blank Role
  sio:000227 rak:C000001 . # isRoleOf

rak:P400001 a rak:0000309 ; # DNA Extraction Process
  sio:000230 rak:C000001 ; # hasInput
  sio:000291 rak:C000001 ; # hasTarget
  sio:000355 rak:R000001 ; # realizes
  sio:000068 rak:B000001 . # isPartOf (Batch)

rak:B000001 a rak:0000308 . # Laboratory Processing Batch
```

Uncertain identity, batch membership and assay applicability are represented as evidence-bearing statements with an explicit confirmed, provisional, unresolved or inapplicable status rather than as ground truth, so a reused or ambiguous label never merges two physical controls. Whole-cell D6300 and purified-DNA D6322 controls enter the workflow at different processes, which lets the graph distinguish extraction and lysis evaluation from PCR, library and sequencing evaluation. The control module contains 489 entities and 3,800 triples, including 51 material-specific role individuals and 47 sequence occurrences.

3.3.10 Archived-soil pH pattern

Archived-soil pH instantiates the generic measurement pattern of Equation 1 but binds it to an archived specimen rather than to a field site. The measuring process consumes and targets an existing specimen, outputs a single pH value, is specified by the $CaCl_2$ protocol, and is part of a dated measurement session. The value is the measurement value of an acidity quality borne by the same specimen.

$$\begin{aligned}\text{pHMeasurementProcess} \sqsubseteq\ & \text{MeasurementProcess} \sqcap \exists \text{hasInput.Specimen} \\ & \sqcap \exists \text{hasTarget.Specimen} \sqcap \exists \text{hasOutput.pHValue}\end{aligned} \tag{43}$$

$$\mathrm{pHValue} \sqsubseteq \mathrm{Quantity} \sqcap \exists \mathrm{isMeasurementValueOf.AcidityQuality} \quad (44)$$

$$\mathrm{AcidityQuality} \sqsubseteq \mathrm{Quality} \sqcap \exists \mathrm{isAttributeOf.Specimen} \quad (45)$$

$$\mathrm{pHMeasurementProcess} \sqsubseteq \exists \mathrm{isPartOf.MeasurementSession} \quad (46)$$

Listing 10 shows a single admitted pH observation. The specimen `rak:6000001` bears the acidity quality `rak:5600001`; the value `rak:4600001` carries the literal `7.8`, has unit UO:0000196, and points back to the quality with `sio:000215` and to its producing process with `sio:000232`. The process is specified by the $CaCl_2$ protocol and is part of the session `rak:B100001`. Rows that fail source, date, specimen or quality-control admission remain in the audit table and are not asserted as measurements (Section 2.8).

Listing 10 Turtle representation of an archived-soil pH measurement.

```
rak:P500001 a sio:001054 ; # Measuring Process
  sio:000230 rak:6000001 ; # hasInput (Specimen)
  sio:000291 rak:6000001 ; # hasTarget
  sio:000229 rak:4600001 ; # hasOutput (pH Value)
  sio:000339 rak:L000020 ; # isSpecifiedBy (CaCl2 Protocol)
  sio:000068 rak:B100001 . # isPartOf (Session)

rak:4600001 a sio:001089 ; # pH Value
  sio:000300 "7.8"^^xsd:double ; # hasValue
  sio:000221 uo:0000196 ; # hasUnit (pH)
  sio:000215 rak:5600001 ; # isMeasurementValueOf
  sio:000232 rak:P500001 . # isOutputOf

rak:5600001 a pato:0001842 ; # Acidity Quality
  sio:000011 rak:6000001 ; # isAttributeOf (Specimen)
  sio:000216 rak:4600001 . # hasMeasurementValue
```

3.4 OWL and RDF implementation

We generated the OWL/RDF implementation using custom Groovy scripts utilizing the OWL API 5 [63]. The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata, quality control metrics, laboratory controls, archived-soil pH, and taxonomic abundances into individual ABox modules [64].

We use OpenLink Virtuoso [65] for RDF storage and query answering. In a development benchmark, a project-specific forward-rule program applied selected RDFS/OWL rules for subclass, subproperty, domain, range, inverse, symmetric, equivalence and transitive-property assertions. The program used three scheduled passes, including only a one-hop transitive pass. It did not demonstrate convergence to a fixed point and did not implement arbitrary OWL property chains. We therefore describe this output as a *selected rule expansion*, not as complete OWL-Horst [66] closure, and we report exact module-level entity counts (Section 4) rather than historical asserted or inferred whole-graph triple counts.

3.5 Schema definition

We used Shape Expressions (ShEx) [57] to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph's structure, ensuring data quality and compliance before loading it into the database.

Our ShEx implementation defines shapes for Sampling Sites, Soil Samples, Environmental Measurements, DNA Extracts, XRF processes, Sequencing Datasets, laboratory controls, and archived-soil pH observations. These shapes guarantee that essential attributes are invariably present. For instance, Listing 11 demonstrates how the schema enforces that every Sampling Site is explicitly typed and labeled, while capturing geographic coordinates and an optional textual description.

Listing 11 ShEx shape for Sampling Site validation.

```
<#SamplingSiteShape> {
    rdf:type [rak:0000002] ; # Must be of type Sampling Site
    rdf:type . * ; # May have other inferred types
    rdfs:label . ; # Must have a label
    dcterms:description xsd:string ? ; # Optional description
    geo:asWKT . ; # Mandatory geographic coordinates
}
```

In words, the shape `<#SamplingSiteShape>` requires that every conforming node carry an explicit type assertion to `rak:0000002` (Sampling Site) and an `rdfs:label`; it permits any number of additional inferred types and optionally allows a free-text description. Coordinates are mandatory in the shape, and all 70 sampling-site individuals in the released module satisfy it. Records that omit the mandatory type, label or geometry are rejected before they reach the triple store, and a deliberately coordinate-free fixture in the negative-fixture suite (Section 5) proves that the rejection fires.

The remaining shapes reuse the same closed-world idiom but encode the domain-specific provenance and measurement contracts that a type assertion alone cannot express. Listing 12 shows the soil-sample shape: a conforming sample must be explicitly typed as a replicate, carry a label and a stable source identifier, and be derived from a sampling site. The derivation edge is what lets a query traverse from a specimen to its site without joining through the sampling event.

Listing 12 ShEx shape for a soil sample.

```
<#SampleShape> {
    rdf:type [sio:001418] ; # Replicate
    rdf:type IRI * ; # Plus a specific material type
    rdfs:label xsd:string ;
    dc:identifier xsd:string ; # Stable source identifier
    sio:000244 IRI ; # isDerivedFrom -> Site
}
```

The DNA-extract and XRF shapes enforce the canonical measurement reification (split quality and value) of Equation 1. Listing 13 requires the concentration value to carry a numeric concentration, the nanograms-per-microlitre unit, and links back to both its quality and its producing measurement process, so a value that is detached from its process or that uses the wrong unit is rejected.

Listing 13 ShEx shape for a DNA-concentration value.

```
<#DNAConcentrationValueShape> {
    rdf:type [rak:0000044] ; # Concentration value
    sio:000215 @<#DNAConcentrationQualityShape> ; # -> Quality
    sio:000232 @<#MeasurementProcessShape> ; # -> Process
    sio:000221 [uo:0000275] ; # hasUnit -> ng/uL
    rak:2000015 xsd:double ; # hasDNAConcentration
}
```

The XRF shape encodes the two acquisition workflows directly: a conforming XRF process must either take a collected specimen as input (the laboratory workflow) or target a recorded sampling site (the field workflow), and it must produce at least one value. This disjunction is what allows one shape to cover both workflows while still rejecting a process that has neither an input nor a target, the defect exercised by one of the negative fixtures.

Listing 14 ShEx shape for an XRF measurement process.

```
<#XRFProcessShape> {
    rdf:type [rak:0000025] ;
    (
        sio:000230 IRI | # lab: hasInput -> specimen
        sio:000291 IRI # field: hasTarget -> site
    ) ;
    sio:000339 IRI ; # isSpecifiedBy -> Protocol
    sio:000229 @<#XRFValueShape> + ; # hasOutput -> 1+ values
}
```

4 Data records

The resource is distributed as a single directory tree in which every payload file is declared in a machine-readable manifest (`PRE_RELEASE_MANIFEST.tsv`) with its byte size, SHA-256 checksum, category, release status and licence gate. The package builder adds `PRE_RELEASE_MANIFEST.tsv` itself explicitly rather than listing its own checksum recursively. Three payloads exceed 256 MiB (`metadata/metagenome/eq.emapper.annotations.gz`, `metadata/taxonomy/feature-table-trips1-5.tsv` and `ontology/rubalkhali_taxonomy_abox.ttl`); the downloadable reviewer archive omits them and lists them with byte size and SHA-256 checksum in a separate bulk manifest, and a complete deposit supplies the byte-identical files. The raw sequencing data are deposited in the European Nucleotide Archive (ENA) under the umbrella project accession `PRJEB104209`, with amplicon sequencing under project `PRJEB106069`. The knowledge graph is accessible via a public SPARQL endpoint at https://rubalkhali.science/sparql and through a web interface at https://rubalkhali.science/, and all source code for data processing, ontology generation, and validation is maintained at https://github.com/bio-ontology-research-group/empty-quarter.

4.1 Knowledge-graph modules

The primary artifacts are the knowledge-graph modules, distributed as OWL/XML and Turtle files and structured by data type (Table 3). The base ontology (`rubalkhali.owl`) holds the terminology; each remaining module holds the instance data for one domain and shares the same site, specimen and process identifiers, so the modules can be loaded together into any RDF store or used individually.

Table 3 Knowledge-graph modules in the deposit.

Module	Content
`rubalkhali.owl`	Base ontology (TBox)
`rubalkhali_sites.owl`	Sampling sites with coordinates and biome annotations
`rubalkhali_samples.owl`	Soil and plant specimen individuals with compartment types
`rubalkhali_measurements.owl`	Field environmental measurements and climate records
`rubalkhali_xrf.owl`	XRF measurement values from the field and laboratory workflows
`rubalkhali_dna.owl`	DNA extracts with concentration data
`rubalkhali_sra.owl`	Sequencing run records linked to ENA accessions
`rubalkhali_qc.owl`	Sequencing quality metrics (GC content, duplicate rate, sequence counts)
`rubalkhali_controls.ttl`	Laboratory-control materials, roles, batches and sequence occurrences
`rubalkhali_ph_eq_ph_shared_v1_0_0.ttl`	Archived-soil pH measurements and sessions
`rubalkhali_taxonomy_abox.ttl`	Taxon abundance observations over the feature profiles

4.2 Tabular data products

Alongside the graph, the deposit distributes the curated tabular data products from which the modules are generated: the versioned source ledger linking site and trip to specimen, extraction, library, sequence run, quality-control decision and derived community; the canonical Trips 1–5 amplicon feature table and its taxonomy mapping; the correction and alias ledgers that record every deterministic metadata repair without overwriting source values (the alias ledger maps the 36 genuine Trip-1-only records of numeric sites 61–64 to the four existing named site individuals); the curated field environmental table; the field and laboratory XRF tables together with the audited chemical-mapping ledger; and the versioned archived-soil pH dataset `EQ-PH-SHARED-v1.0.0`. The canonical feature table holds 1,271 profiles, of which the ecological analysis retains 1,237 (1,227 profiles from the primary frame of sites 1–60); the 34 excluded profiles are the 24 controls, nine biological profiles below the declared 1,000-read ecological threshold and one additional `T1Dr1` run with 934 reads. The four derived companion-analysis inputs are distributed as records for downstream statistical re-execution rather than as graph modules: the paired PMA table (`metadata/relic-dna/PMA_ASV_table.tsv`; 18 analytical columns from nine aliquot pairs, PMA-treated and untreated, with controls kept separately), the 150 CoverM profiles (`metadata/metagenome/coverm_profiles.tar.gz`), the eggNOG annotations (`metadata/metagenome/eq.emapper.annotations.gz`) behind the 990-genome analytical subset, and the measured-function input archive (`metadata/metagenome/measured_function_inputs.tar.gz`) with the six exact source tables of the genome-derived versus PICRUSt2 comparison. These inputs do not reproduce raw shotgun or PMA read processing, assembly, binning, dereplication, annotation or profile generation.

4.3 Validation schemas, queries and evidence

The deposit includes the artifacts needed to re-validate the resource: the ShEx shapes that define the structural contract for each entity family, the positive and negative structural fixtures, the archived competency SPARQL queries, and the machine-readable evidence bundles (reconciliation summaries, audit tables, validation records and checksum manifests) produced by the release workflow. A companion data dictionary (`metadata/DATA_DICTIONARY.tsv`) records the field names, data types, units, missing-value conventions and definitions for the curated canonical tabular records. Quantities carry units only where a unit is documented at the source: environmental source tables retain degrees Celsius, hectopascals (millibars) and percent relative humidity; in the graph, pressure magnitudes are multiplied by 100 and asserted in pascals (`obo:UO_0000110`), climate records use degrees Celsius and millimetres, and DNA concentrations are in nanograms per microlitre. XRF concentrations carry no documented unit assertion and must not be treated as percentages.

The data and source code to generate the Rub' al-Khali knowledge graph is available at .

Add Zenodo DOI URL here

5 Technical validation

We validated the knowledge graph using four complementary approaches, each targeting a distinct aspect of data quality: competency question evaluation via SPARQL queries, structural validation using Shape Expressions (ShEx), logical consistency checking with the ELK reasoner, and usability assessment through a web portal.

5.1 Competency questions and SPARQL queries

To validate whether the full knowledge graph can be used to answer queries about the Rub' al-Khali, we defined competency questions expressed as SPARQL queries [67] with known expected answers. We automated this validation process to enable regression testing during ontology updates. The validated competency questions include retrieving XRF measurements, listing sampling sites with coordinates, extracting DNA extraction data, tracing provenance from soil samples to sequencing runs, and querying taxonomic composition. The competency questions are designed to test questions that are needed for operating the website, and to support different types of queries within our dataset.

Table 4 lists the competency questions we predefined for validation. Each is expressed as a SPARQL query with a known expected answer and is run as a regression test whenever the knowledge graph is regenerated. Together they cover every core module: geochemistry (CQ1, CQ4), the site catalogue and its biome grounding (CQ2, CQ3), the molecular provenance chain (CQ5, CQ6), and taxonomic composition (CQ7).

Table 4 Predefined competency questions used for validation.

ID	Competency question
CQ1	What are the XRF analyte concentrations measured at a given sampling site?
CQ2	Which sampling sites are recorded, and what are their GPS coordinates?
CQ3	Which biome and environmental features are assigned to each sampling site?
CQ4	What are the Light Elements percentages across the XRF measurements?
CQ5	Which DNA extraction records (extract, concentration, protocol) exist for a given sample?
CQ6	What is the full provenance chain from a soil sample to its sequencing run?
CQ7	What is the taxonomic composition and abundance of a given sample or run?

We show several example SPARQL queries. Listing 15 retrieves XRF measurements for a specific site, requiring traversal of the measurement process, quality, and value entities.

Listing 15 SPARQL query to retrieve field XRF measurements for Site 10. The query starts from the field-XRF measuring processes whose target is the site and walks to the individual analyte concentrations.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?processLabel ?siteLabel ?analyte ?concentration
WHERE {
  ?process a rak:0000025 ;
           rdfs:label ?processLabel ;
           sio:000291 ?site ;
           sio:000229 ?value .
  FILTER(CONTAINS(LCASE(STR(?processLabel)), "field xrf"))
  ?site rdfs:label ?siteLabel .
  FILTER(?siteLabel = "Site 10")
  ?value sio:000215 ?quality ;
         rak:2000012 ?concentration .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000032)
}
ORDER BY ?processLabel ?analyte
```

Read in prose, the query first selects every field-XRF measuring process whose target site is labelled `Site 10`, then walks from each output value of that process to the quality it measures and to the recorded concentration. Field XRF is linked to the site rather than to a specimen, so no sample appears in the pattern, and no unit is bound because the XRF exports document none. The final filter excludes the Light Elements pseudo-analyte (`rak:0000032`) so that only individually quantified elements appear in the result, and the query returns one row per process and analyte combination.

The release workflow executes this exact query (`sparql/field_xrf_site10.rq`) against the checksummed base, site and XRF modules and requires the fixed expected cardinality of 46 analyte-value rows; it returned exactly 46 rows for Site 10, 24 from field process Test 5847 and 22 from Test 5848, and the result table and validation record are archived under `evidence/competency-query/`. No broader competency-query pass rate is claimed: only this query carries a fixed expected cardinality in the release workflow.

Listing 16 retrieves taxonomic abundance profiles, joining across bioinformatic workflows, FASTQ datasets, and taxon observations.

Listing 16 SPARQL query to retrieve taxonomic composition and abundance for all samples.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?run ?taxonLabel ?count ?relativeAbundance
WHERE {
  ?proc a rak:0000071 ;
        sio:000230 ?fastq ;
        sio:000229 ?dsAbs , ?dsRel .
  ?dsAbs a rak:0000074 .
  ?dsRel a rak:0000075 .
  ?fastq rdfs:label ?runLabel .
  BIND(REPLACE(?runLabel, "FASTQ dataset for ", "") AS ?run)
  ?dsAbs sio:000059 ?valAbs .
  ?valAbs rak:2000026 ?count .
  ?valAbs sio:000215 ?qualAbs .
  ?qualAbs sio:000011 ?taxon .
  ?taxon rdfs:label ?taxonLabel .
  ?qualRel sio:000011 ?taxon ;
           a rak:0000072 ;
           sio:000216 ?valRel .
  ?valRel rak:2000020 ?relativeAbundance .
  FILTER(?taxon != ?fastq)
}
ORDER BY ?run DESC(?count) LIMIT 100
```

In words, the query identifies every bioinformatic workflow that produces both an absolute and a relative taxonomic dataset for the same FASTQ input, then for each absolute observation it retrieves the read count, the associated taxon label, and the matching relative abundance from the parallel relative dataset. Results are grouped by sequencing run and ordered by descending count, so the most abundant taxa per run appear first.

We added an additional query (Listing 17) to retrieve specific taxa abundances alongside environmental constraints, demonstrating the ability to cross-reference biological and environmental properties.

Listing 17 SPARQL query to retrieve Pseudomonas relative abundance at sites with high temperature.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX pato: <http://purl.obolibrary.org/obo/PATO_>
PREFIX ncbitaxon: <http://purl.obolibrary.org/obo/NCBITaxon_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?siteLabel ?temp ?relAbundance
WHERE {
  ?site a rak:0000002 ;
        rdfs:label ?siteLabel .

  # Environmental Temperature (site hasAttribute quality)
  ?site sio:000008 ?tempQual .
  ?tempQual a pato:0000146 ;
            sio:000216 ?tempVal .
  ?tempVal rak:2000003 ?temp .
  FILTER(?temp > 35.0)

  # Derived Sample and Taxon Abundance
  ?sample sio:000244 ?site .
  ?ext a rak:0000309 ; sio:000230 ?sample ; sio:000229 ?dna .
  ?libPrep a rak:0000065 ; sio:000230 ?dna ; sio:000229 ?lib .
  ?seq a rak:0000066 ; sio:000230 ?lib ; sio:000229 ?fastq .

  ?bioinf a rak:0000071 ;
          sio:000230 ?fastq ;
          sio:000229 ?dsRel .
  ?dsRel a rak:0000075 ;
         sio:000059 ?valRel .
  ?valRel rak:2000020 ?relAbundance ;
          sio:000215 ?qualRel .
  ?qualRel sio:000011 ncbitaxon:286 . # Pseudomonas
}
```

In words, the query first selects every sampling site at which the recorded environmental temperature exceeds 35 °C, then for each such site it walks the full provenance chain — soil sample, DNA extract, library preparation, sequencing run, FASTQ dataset, bioinformatic workflow, relative abundance dataset — down to the relative abundance value of *Pseudomonas* (`ncbitaxon:286`). Each row of the result therefore reports a single hot site together with its corresponding *Pseudomonas* relative abundance, demonstrating a join that crosses the environmental, geochemical, and biological domains in a single SPARQL query.

5.2 Source reconciliation and measurement validation

The executed evidence workflow produced a row-level sample ledger and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. A site-alias gate independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement, accounting for all 36 affected source rows.

The environmental-metadata gate reproduced 274 nonblank source records and verified all 24 exact-cell correction-ledger dispositions against the corresponding raw cells. All retained field temperatures, pressures and relative humidities fell within their declared physical ranges; the unresolved Trip 5 site 40 value of 194% was preserved in the ledger but is absent from both the curated table and the environmental RDF module. The XRF provenance audit reproduced all 13,390 reported Trips 1–4 cells under the maximum-positive rule and all 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells, and verified that the canonical laboratory input comprises all 725 processed records. An independent chemical-identifier gate checked all 93 instrument channels against the pinned ChEBI release 247 and the checksummed PubChem snapshot of 23 July 2026 (Section 2.7); all rows passed, and audited nulls are retained for the unmapped LE pseudo-analyte and unsupported oxides rather than being replaced by lexical matches. The pH reconciliation independently accounts for all 1,168 workbook rows, matches every admitted row to exactly one existing specimen, and reproduces the five dispositions reported in Section 4.

5.3 ShEx schema validation

We ran the ShEx validation described in Section 2 (shapes defined in Section 3.5) over every regenerated ABox module before loading into Virtuoso. The tractable modules passed their assigned ShEx shapes. The pH module was validated separately over all 712 measurement processes, values and acidity qualities and all 29 sessions, and the laboratory-control module passed its shape and scientific-invariant checks. The ShEx suite also tests rejection, not only conformance: negative fixtures that differ from a valid instance in exactly one respect (a sampling site without coordinates, a value with an invalid datatype, a reversed value-to-quality direction, and an XRF process with neither an input nor a target) are each rejected by the released shapes, so a passing negative test identifies the constraint that caught it. The multi-gigabyte taxonomy ABox remains outside this memory-bounded run; it is instead parsed in full by an independent Turtle parser and scanned by a streaming structural validator, which accounted for 44,528,482 triples in 5,953,690 subject blocks with zero structural violations. This establishes full-file syntax and generator-specific structural validation, not a claim that the taxonomy ABox passed the unexecuted ShEx shapes. The Turtle listings of Section 3 are checked mechanically as well: `scripts/validation/verify_manuscript_listings.py` parses each listing and resolves every printed triple and every prose IRI against the staged modules, so a drifted listing fails the workflow instead of reaching a reader as an unusable pattern.

5.4 Logical consistency

The ELK consistency check described in Section 2 was re-run whenever the TBox changed or a new ABox module was introduced. A clean rerun on the current tractable modules loaded 781,293 axioms from 10 distinct ontology IRIs and reported the resulting merged ontology as consistent; the generated ecosystem and laboratory-control modules were included. The run excluded the multi-gigabyte taxonomy ABox and large reference modules and did not precompute the class hierarchy, so it establishes bounded consistency of the included modules rather than whole-graph OWL 2 DL consistency. ELK is incomplete for universal restrictions, inverse properties and other constructs used by imported modules, and the excluded ABoxes were not covered; likewise, the selected forward-rule program of Section 3.4 has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure [66]. A separate label gate merged 782,229 triples from the 11 current files and confirmed exactly one label for each of 104,697 labelled `RAK_` subjects; the log, input hashes and validator hashes of both gates are archived under `evidence/semantic-validation/`.

5.5 Compute-time metrics

Table 5 summarizes the computational cost of the key deployment steps of the reference Virtuoso deployment. The selected forward-rule expansion (three passes; see Section 3.4) runs in approximately 114 minutes. ELK consistency checking over the asserted modules takes under 2 seconds. Competency question latency on the Virtuoso endpoint ranges from 5 ms for simple site look-ups (CQ2) to over 1 second for queries that traverse inverse properties across the entire DNA extraction workflow (CQ5).

Table 5 Compute-time metrics for key deployment and validation steps.

Step	**Time**
RDF module loading (Virtuoso, 9 ABox files)	140 s
Selected rule expansion (3 passes)	114 min
ELK classification (asserted only)	1.6 s
CQ1 – XRF measurements, Site 10	23 ms
CQ2 – Sampling sites + GPS	5 ms
CQ3 – Biomes per site	5 ms
CQ4 – Light Elements percentages	8 ms
CQ5 – DNA extraction records	1,342 ms
CQ6 – Provenance chain	255 ms
CQ7 – Taxonomic composition	>300 s

The taxonomy composition query (CQ7) exceeds a 5-minute timeout because it joins absolute and relative abundance datasets across 1.4 million observations. This query is not used by the web portal; the portal retrieves taxonomic data per sample rather than across all runs, which completes in sub-second time.

5.6 Usability validation (web portal)

We developed a web portal (https://rubalkhali.science/) to provide user-friendly access to the data. The application architecture provides an end-to-end validation of the knowledge graph: the backend dynamically generates SPARQL queries to fetch content directly from the knowledge graph in real time without maintaining a separate relational database. Every page rendered constitutes a functional test of the underlying data. This architecture ensures that the website always reflects the current state of the knowledge graph, and any data integrity problem manifests as a visible error on the portal.

5.7 FAIR compliance

The design decisions described in the preceding sections collectively implement the FAIR Data Principles [30]. We describe how the resource addresses each principle in turn.

To make the data *findable*, we assigned a globally unique and persistent IRI under the `https://rubalkhali.science/kb/` namespace to every named individual, so that each site, specimen, process, value and dataset is directly and stably addressable rather than identified only by a row in a table. The sequencing data are registered with the European Nucleotide Archive under the umbrella accession PRJEB104209, and the resource is described with rich metadata: the machine-readable manifest indexes every payload file, and the data dictionary defines every curated field. Because the knowledge graph encodes its own terminology, the metadata and the data share one namespace and can be retrieved together.

To make the data *accessible*, we expose the knowledge graph through a standard SPARQL 1.1 endpoint and a web interface, both of which use open, universally implementable protocols and require no authentication for read access. The raw sequences are retrievable from ENA through their accessions, and the deposited modules can be downloaded and loaded into any RDF store, so access does not depend on the continued operation of our endpoint.

To make the data *interoperable*, we represent all content with established shared vocabularies: SIO as the upper-level framework, ENVO for environmental context, ChEBI for chemical entities, PATO for qualities, UO for units, and NCBI Taxonomy for taxa. The graph uses the standard formal languages OWL 2 DL, RDF/Turtle and SPARQL, and because domain entities are grounded in these widely used ontologies, the resource can be joined with other datasets that use the same vocabularies. The qualified references between modules (for example from a specimen to its site or from a value to its producing process) are themselves asserted as RDF, so the cross-references that give the data their meaning travel with the data.

To make the data *reusable*, we describe the resource with detailed and accurate provenance that records the complete chain from physical sample collection through DNA extraction, library preparation, sequencing and quality control to the derived community profiles, together with the correction and alias ledgers that document every deterministic repair to the source records. The data are released under CC-BY 4.0, the generation and validation code is open-sourced, and the release is versioned and archived so that a cited version remains stable. These attributes let a reuser understand the origin and limitations of each observation and integrate the resource into new work with confidence.

6 Usage notes

The knowledge graph is publicly accessible via the SPARQL endpoint at https://rubalkhali.science/sparql. Users can submit SPARQL 1.1 queries using standard clients (e.g., `curl`, Python's `SPARQLWrapper`, R's `SPARQL` package). The web portal at https://rubalkhali.science/ provides a graphical interface for browsing sites, samples, and measurements without requiring SPARQL expertise.

The semantic structure of the knowledge graph enables cross-domain queries. For example, researchers investigating the relationship between soil geochemistry and microbial diversity can construct a single query that retrieves calcium concentrations alongside Shannon diversity indices computed from taxon abundances. Executing this query in a relational database would necessitate joining at least four separate tables.

The modular OWL architecture permits extending the knowledge graph with new data types, such as metabolomics or remote sensing data. Users can introduce new classes as subclasses of existing SIO entities and generate corresponding ABox modules. We automate the entire lifecycle—from data generation through validation to database loading—using the provided deployment workflow (`deploy_onto.sh`).

6.1 Use cases

Two worked use cases illustrate how the graph supports analyses that its individual source tables do not, following the practice of resource papers such as Tara Oceans [29] and MetagenomicKG [35] of demonstrating utility on concrete questions.

Use case 1: environmental context for a community observation.

A reuser studying thermotolerance wants the bacterial genera observed at the hottest sampled sites, together with the geochemical and climate context of those sites. Because field measurements, XRF sessions, climate records and taxon observations all instantiate the same measurement idiom around the same site individuals (Section 3), this is a single SPARQL query rather than a chain of table joins keyed on inconsistent site labels: Listing 17 retrieves *Pseudomonas* relative abundance at sites whose recorded field temperature exceeds a threshold, and the same pattern extended with the geochemistry fragment of Listing 15 adds, for example, calcium concentrations or monthly precipitation for the returned sites. The competency questions of Table 4 are regression-tested variants of exactly these traversals, so the query routes shown here are re-validated whenever the graph is rebuilt. Of these routes, the field-XRF retrieval of Listing 15 is workflow-validated at a fixed cardinality of 46 rows (Section 5).

Use case 2: batch-aware contamination sensitivity.

The laboratory-control pattern (Section 3) links every extraction blank to its extraction batch, and every biological profile to the batch that produced it. The companion ecology analysis [43] consumed these frozen links directly: the 351 candidate contaminant ASVs identified from the Trip 5 blanks were removed only from the 217 batch-linked Trip 5 profiles (and the corresponding Trip 4 screen only from its linked profiles), never from unlinked ones, and the analysis then re-evaluated its tracked headline conclusions on the filtered table; every verdict was retained. The candidate tables, filtered-table builder and per-metric outcomes are archived as evidence bundles in the reproducibility repository, so a reuser can repeat the sensitivity analysis under a different candidate set or link stringency without re-deriving the batch structure from laboratory records.

6.2 Design rationale

We adopted the Semanticscience Integrated Ontology (SIO) as the upper-level framework because it offers a small set of relations (`hasInput`/`hasOutput`, `hasTarget`, `hasMember`, `isMeasurementValueOf`, `hasValue`) that generalize across sampling, measurement, and provenance patterns. Sample collection, XRF analysis, DNA extraction, sequencing, and taxonomic classification are all instantiations of the same `Process--hasInput/hasOutput--Object` idiom; quantitative results are uniformly represented as `Quantity isMeasurementValueOf Quality` pairs, with a unit asserted where the source documents one; XRF concentration values have no unit assertion because the XRF exports document none. This uniformity enables cross-domain SPARQL queries that traverse from a climate observation through a sampling site to a taxon abundance without per-domain join logic. Recent work on simplified upper-level ontologies, such as SULO [68], takes this argument further by strengthening fundamental relations and reducing reliance on specialized predicates; our patterns are compatible with this direction.

We chose ShEx for ingest validation because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL is an obvious alternative supported in-database by several triple stores; a future port of the shapes from ShEx to SHACL would enable in-database validation at load time. We deployed Virtuoso as the triple store because it is available as free software under the GNU General Public License, supports OWL-Horst forward-chaining materialization, and has demonstrated operational maturity in large-scale linked-data deployments.

Full OWL 2 DL reasoning over the combined TBox and the ABox is not computationally feasible with HermiT on commodity hardware. We therefore use ELK [58] to validate the logical consistency of the knowledge graph. For query answering, we rely on a selected forward-rule expansion in Virtuoso, which physically expands the graph by applying subclass, subproperty, inverse, symmetric, equivalence, and transitivity rules; this expansion has not been demonstrated to reach a fixed point and is not complete OWL-Horst closure [66]. Two decades of work on scaling OWL reasoning—materialization-based approaches [66], query rewriting, modular reasoning, and consequence-based calculi such as ELK itself—provide several alternative points in the design space, and a more expressive reasoner could be applied to a sub-module of the TBox in future work.

6.3 Limitations and outstanding challenges

Several challenges remain open. First, the assay-aware control screen identifies 351 of 351,472 ASVs using 17 extraction-day blanks with batch links to Trip 5 profiles. Removing these candidates only from the 217 co-extracted Trip 5 biological profiles removes 2.19 % of reads in aggregate, and the unfiltered table remains canonical. Inference is bounded because the design provides only one blank per extraction batch, concentration data are absent, Trip 4 has no positive control and has an unsequenced blank affecting all or part of 23 sites, and Trip 5 has no 16S positive control. A reuser should use the control module and audit tables to select compatible controls by assay, stage and batch, and should not treat contamination as eliminated. Second, archived-soil pH is available for 712 admitted specimens under the specified $CaCl_2$ protocol, but coverage is incomplete and non-random and should not be interpreted as an in-situ field map. Third, the XRF sources lack machine-readable unit and non-detect semantics, and Trips 1–4 and Trip 5 use disclosed but different repeat-formula rules; field and laboratory XRF values cannot be assumed interchangeable, and their current comparison is site-level rather than same-aliquot. Fourth, more complex reasoning—for example over disjunctions arising from ENVO biome classifications or negative axioms about taxon absence—would require either a different reasoner or the extraction of a sub-module amenable to a more expressive calculus. Finally, the resource depends on upstream ontologies (SIO, ChEBI, ENVO, NCBI Taxonomy) that evolve independently; keeping the deployed graph aligned with their releases is an ongoing maintenance task and a known reproducibility risk for any SPARQL query that bottoms out in a specific external IRI.

6.4 Sustainability and updates

The dataset is versioned and archived on Zenodo, and the deployment workflow is reproducible from the public repository. We intend to extend the resource as additional field campaigns are processed: new trips will be added as additional ABox modules without changes to the upper-level patterns. Upstream ontology updates will be tracked at scheduled intervals; re-materialization is scripted and runs as part of the deployment workflow, so an upstream release can be absorbed by re-running the workflow. Where an upstream change would alter the semantics of an existing query, we will record the affected competency questions in the changelog so that downstream users can audit the impact.

7 Data availability

Sequencing data for this project are deposited in the European Nucleotide Archive (ENA) under accession `PRJEB104209`, an umbrella study ("The genome of the Rub' al Khali (Empty Quarter)").

8 Code availability

All source code for data processing, ontology engineering, RDF generation, validation, and the web portal is openly available at https://github.com/bio-ontology-research-group/empty-quarter. The repository includes Groovy scripts utilizing the OWL API 5 for OWL/RDF generation, Python scripts for data processing and statistical analysis, ShEx shape definitions for structural validation, SPARQL competency queries, React/FastAPI web application source code and Docker Compose configurations for deployment. The manuscript, RDF generators, source metadata and reproducibility workflow are maintained in the companion repository at https://github.com/bio-ontology-research-group/empty-quarter-ecology-data, which holds the `main.nf` entry point, execution profiles, workflow tests, a pinned Conda recipe and a hash-locked Python environment.

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Competing interests

The authors declare no competing interests.

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